🛍️ Black Friday Purchase Prediction System

End-to-End Machine Learning Pipeline with Deployment

Author: N Lokesh Reddy
Project Type: Regression (Supervised Learning)
Model Used: Bagging Regressor (Decision Tree Base Estimator)

1. Summary

This project presents a comprehensive end-to-end machine learning solution designed to predict customer purchase amounts during high-volume retail events such as Black Friday. Leveraging a dataset of over 550,000 transactional records, a structured analytical pipeline was developed covering data cleaning, exploratory data analysis, outlier treatment, feature engineering, model benchmarking, and deployment.

Multiple regression models were evaluated, and the final selected architecture—a **Bagging Regressor utilizing a Decision Tree base estimator**—demonstrated superior predictive performance. The complete preprocessing and modeling workflow was integrated into a unified Scikit-Learn Pipeline and deployed as a real-time Streamlit web application, enabling business users to generate instant purchase forecasts.

This project demonstrates not only technical modeling capability but also the ability to translate predictive analytics into an operational retail intelligence tool.

2. Project Objective & Workflow

Objective

The primary objective of this project is to predict the continuous target variable **Purchase**, representing the total spending amount for a specific customer-product interaction.

Unlike classification tasks, this regression problem aims to estimate precise transaction values, making it highly valuable for revenue forecasting, demand planning, and customer segmentation.

Structured Workflow

The project followed a systematic machine learning lifecycle:

1. Data Ingestion

Loading and validating the raw dataset (blackFriday_train.csv) containing transactional and demographic information.

2. Data Preprocessing

Cleaning and auditing data integrity to ensure model-ready input.

3. Exploratory Data Analysis (EDA)

Investigating statistical patterns and purchase drivers.

4. Feature Engineering

Transforming raw features into meaningful behavioral predictors.

5. Modeling & Benchmarking

Training multiple regression algorithms and selecting the optimal model through performance comparison.

3. Phase 1: Data Cleaning & Integrity

Data cleaning plays a foundational role in ensuring reliable model performance. The following preprocessing steps were executed:

- **Handling Missing Values:**
  Missing entries in Product_Category_2 and Product_Category_3 were replaced with 0, representing absence of a secondary or tertiary category.
- **Formatting Corrections:**
  The Stay_In_Current_City_Years column contained special characters (e.g., “4+”). These were cleaned and standardized to numeric format.
- **Feature Pruning:**
  User_ID and Product_ID were removed to prevent data leakage and overfitting. These identifiers have high cardinality and no predictive generalization value.
- **Duplicate Auditing:**
  The dataset was examined for duplicate records and cleaned accordingly to maintain integrity.

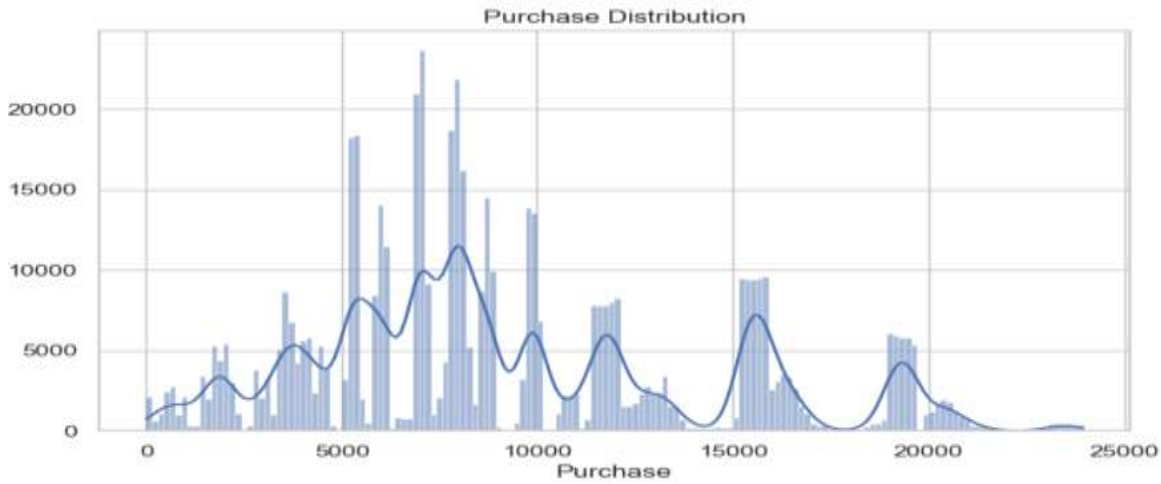
This phase ensured the dataset was statistically consistent and modeling-ready.

4. Phase 2: Exploratory Data Analysis (EDA)

EDA enabled deeper understanding of purchasing behavior beyond raw numeric patterns.

Target Distribution

The Purchase variable exhibited right-skewness, indicating the presence of high-value transactions that could potentially distort regression performance if untreated.



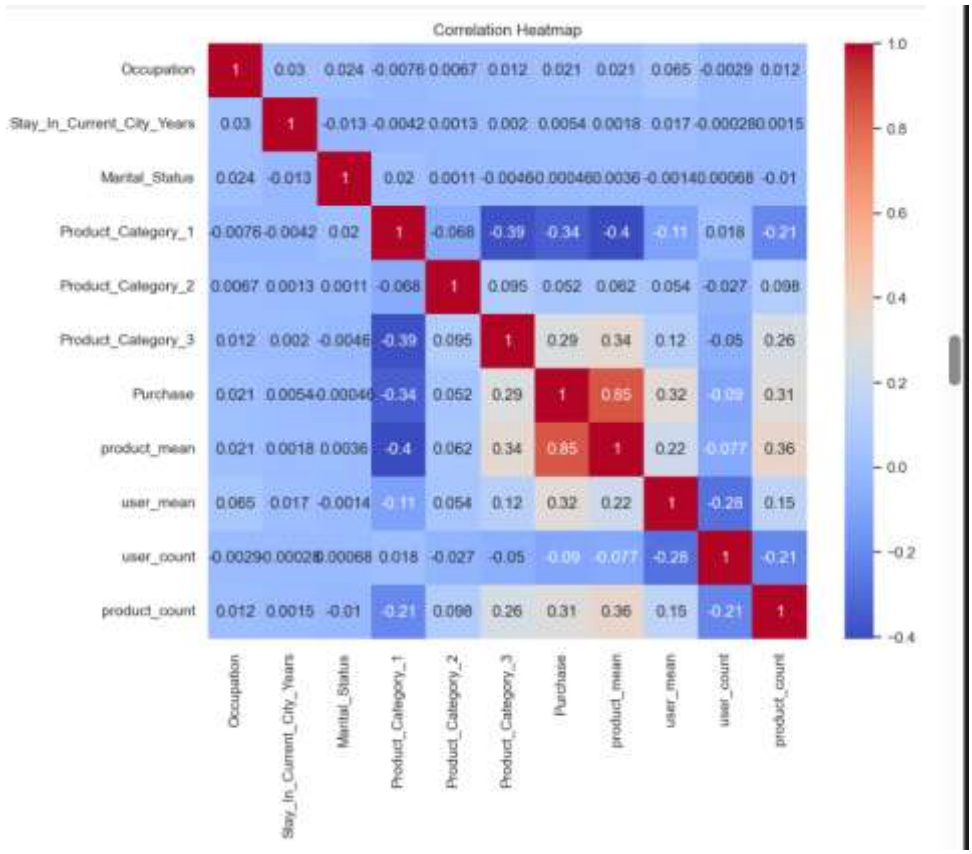
Demographic Drivers

Purchasing behavior was analyzed across:

Gender, Age groups, City Category, Marital Status, Occupation

These analyses revealed distinct spending patterns across demographic segments, suggesting that behavioral and socio-economic factors influence purchase amounts.

Correlation Analysis



A heatmap was generated to identify feature relationships and potential multi-collinearity risks prior to modeling. This step ensured stable model performance and informed feature engineering decisions.

5. Phase 3: Outlier Detection & Strategic Handling

Outliers were detected using boxplots and statistical distribution analysis.

Instead of blindly removing extreme values, the **Interquartile Range (IQR) method** was used to calculate lower and upper bounds.

Calculated boundaries:

- Lower Bound: -3523.5
- Upper Bound: 21400.5

A strategic **capping technique** was applied to limit extreme values within these thresholds.

This approach preserves valuable high-spending behavioral signals while preventing extreme anomalies from disproportionately influencing regression algorithms.

6. Phase 4: Feature Engineering

Feature engineering significantly strengthened model performance by adding contextual intelligence.

Aggregation-Based Feature Creation

To provide behavioral context:

- product_mean → Average purchase per product
- user_mean → Average purchase per customer
- user_count → Total transactions per customer
- product_count → Frequency of product purchase

These features allow the model to capture:

- Customer spending tendencies
- Product popularity trends
- Historical behavioral patterns

This aggregation strategy transforms raw transactional data into predictive intelligence.

Encoding Techniques

- Label Encoding applied to Age and Stay_In_Current_City_Years
- Binary encoding applied to Gender
- One-Hot Encoding applied to City_Category
- First dummy column dropped to avoid the Dummy Variable Trap

Standard Scaling

StandardScaler was applied to normalize feature distributions.

This ensures that features with large numeric ranges (e.g., aggregated means) do not dominate model calculations, particularly in algorithms sensitive to scale.

7. Phase 5: Model Evaluation & Results

A comprehensive benchmarking approach was used to evaluate multiple regression models:

- Linear Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- AdaBoost
- Gradient Boosting
- Bagging Regressor

Models were evaluated using:

- R^2 Score
- Root Mean Squared Error (RMSE)

Final Model Selection

The best-performing architecture was:

Bagging Regressor

- Base Estimator: DecisionTreeRegressor(max_depth=16)
- Number of Estimators: 100

This configuration provided optimal variance reduction while maintaining strong nonlinear modeling capability.

Pipeline Integration

The preprocessing steps (scaling + encoding) and final model were integrated into a unified Scikit-Learn Pipeline.

The pipeline was exported as:

Black_Friday_Model.h5

This ensures seamless and consistent deployment without preprocessing mismatch.

8. Deployment: Streamlit Application

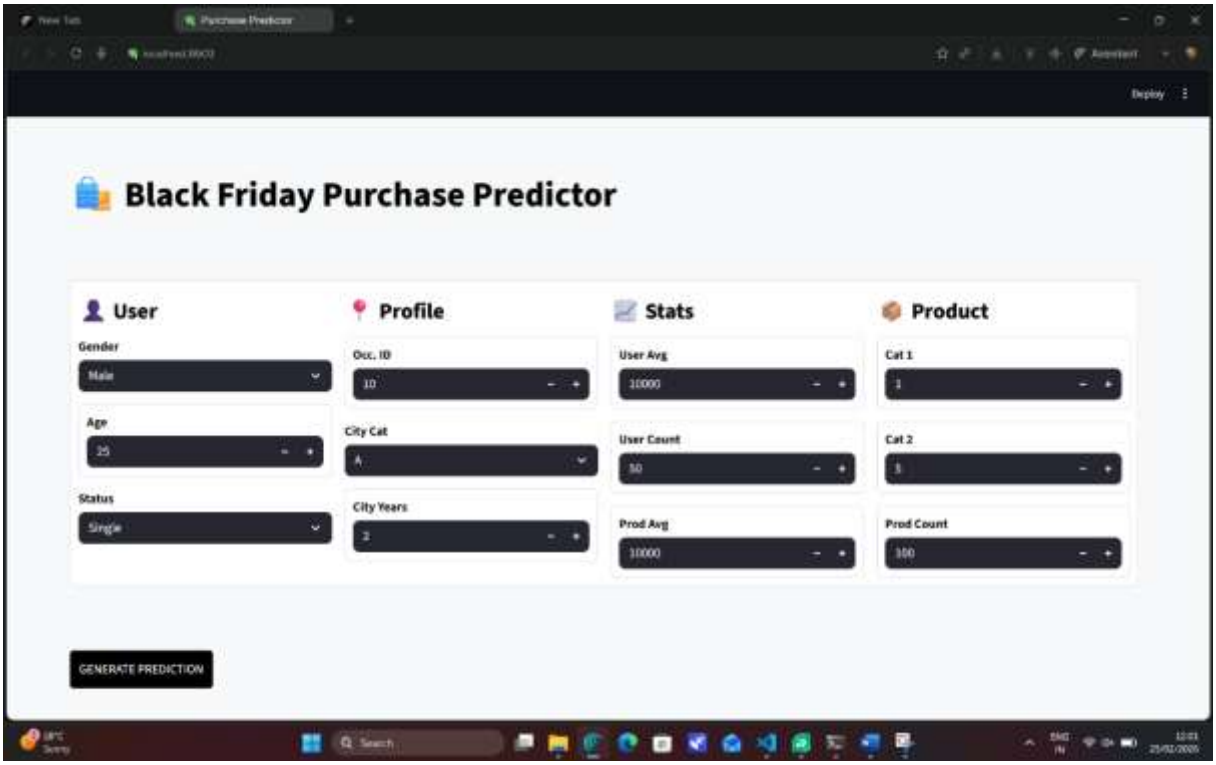
The trained model was deployed as a real-time Streamlit dashboard.

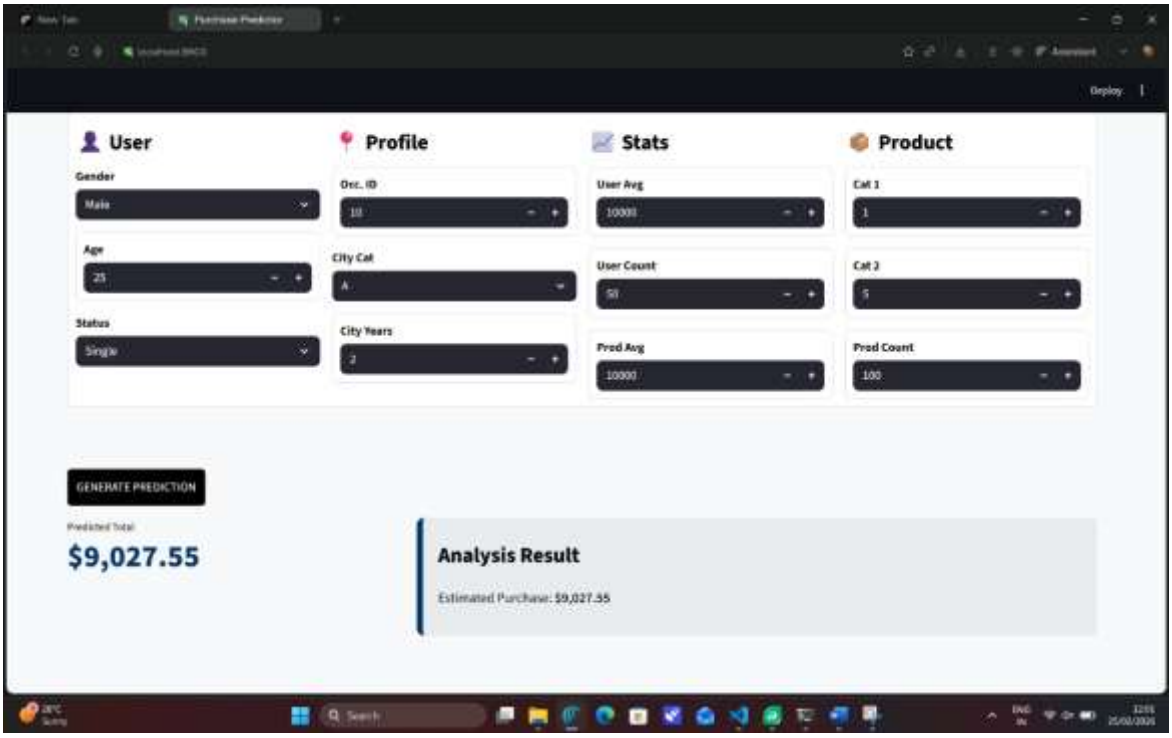
The application enables business users to:

- Input customer demographic attributes
- Provide behavioral statistics
- Select product characteristics
- Instantly receive predicted purchase amounts

The model is loaded using joblib with caching to optimize performance and reduce inference latency.

9. Application Interface





The dashboard is divided into structured sections:

- User Profile
- Demographics
- Statistical Indicators
- Product Usage

This structured design improves usability and aligns with how retail analysts assess customer data.

Once inputs are provided, the system displays the predicted purchase amount in a prominently styled metric card for easy interpretation.

10. Business Impact & Strategic Recommendations

The deployed predictor converts predictive modeling into operational retail intelligence.

Strategic Applications:

- Optimize inventory allocation for high-demand product categories
- Forecast revenue by demographic segment
- Identify high-value customer segments
- Improve targeted promotional planning
- Estimate real-time expected transaction values

This solution bridges analytics and retail operations, enabling data-driven strategic decisions during peak sales events.

11. Final Conclusion

This project demonstrates a complete machine learning lifecycle:

- Data preprocessing
- Outlier handling
- Advanced feature engineering
- Regression benchmarking
- Ensemble learning implementation
- Pipeline integration
- Real-time deployment

By combining Bagging Regressor with engineered behavioral features and an interactive dashboard, the system delivers robust and practical purchase predictions suitable for high-volume retail environments.